



THE CHEAT SHEET

60+

AI Agent Concepts

Every term you need to build & ship agents in 2026.

Swipe to test yourself

agentic AI

LLM tools

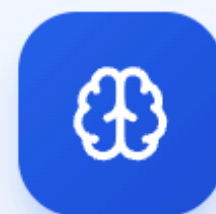
multi-agent

production ML



Chandra Sekhar

AI Engineer & Trainer



How Agents Remember

8

In-context memory

Info kept directly in the LLM's active context window. Fast to access, but limited and lost between sessions.

9

External memory

Persistent storage outside the model — vector DBs for semantic retrieval, SQL for structured data, Redis for fast lookup.

10

Episodic memory

Stored record of past agent interactions, decisions, and outcomes — lets agents learn from prior runs.

11

Semantic memory

General facts and knowledge stored independently of any session — like knowing Python is a programming language.

12

Working memory

Active scratchpad holding intermediate results and notes for the current task only — wiped when the session ends.

13

Memory retrieval

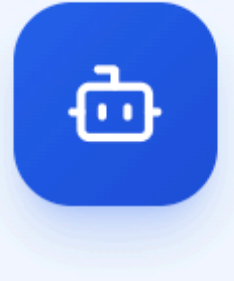
Pulling relevant past memories at query time using semantic similarity — the agent's version of remembering context.

14

Memory consolidation

Summarizing and archiving important past interactions into long-term storage so the agent can reference them later.





What is an Agent?

1

AI Agent

A system that perceives its environment, reasons about it, and takes actions to achieve a goal — without needing a human in every step.

2

Autonomy

Ability to operate without constant human input. The more autonomous an agent, the higher the risk it needs to be managed.

3

Perception

How an agent receives input — text queries, tool call results, images, or any data it can act on next.

4

Action Space

Every possible action the agent can perform — calling tools, running code, querying APIs, browsing web, writing files.

5

Goal

The end objective the agent is trying to reach. All planning, tool use, and reasoning is in service of this goal.

6

Policy

The decision-making strategy that maps what the agent observes to what it does next — the core of agent behavior.

7

Environment

Everything external the agent can interact with — the web, files, databases, APIs, other agents, and the user.





How Agents Think & Plan

15

ReAct

Reason + Act: agent thinks about what to do, takes an action, observes the result, then reasons again. The dominant pattern in production.

16

Chain-of-Thought

Prompting the agent to reason step by step before committing to an answer — dramatically boosts accuracy on hard tasks.

17

Tree of Thought

Generating multiple reasoning branches in parallel, evaluating each, and selecting the best — useful for math and planning.

18

Task decomposition

Splitting a complex objective into smaller, manageable subtasks — each is clearer and easier for the agent to solve.

19

Subgoal

A stepping-stone goal on the way to the final objective. Subgoals give the agent clear checkpoints and reduce errors.

20

Backtracking

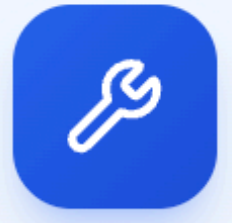
When a path fails, the agent undoes that step and tries an alternative — prevents getting stuck in dead-end loops.

21

Self-reflection

The agent reviewing and critiquing its own output before moving on — catches errors early before they compound.





What Agents Can Do

22

Tool use

Invoking external functions like web search, calculators, or APIs to get real data — the main way agents extend beyond text.

23

Function calling

A formal schema that lets LLMs invoke tools with typed inputs and predictable return values — the standard wiring.

24

Web search

Querying search engines for current, live information — prevents the agent from hallucinating outdated facts.

25

Code execution

Running Python, SQL, or bash scripts inside the agent loop — enables precise computation and file system automation.

26

RAG

Retrieval Augmented Generation: agent fetches relevant docs from a vector DB, injects them, then generates a grounded answer.

27

API calls

Calling external services programmatically — Slack for messages, GitHub for code, Notion for docs, Stripe for payments.

28

Browser use

Controlling a real browser — navigating pages, clicking buttons, filling forms — agents interact like a human would.





How Agents Coordinate

29

Orchestrator

The controller that decides which agent or tool runs next — the brain of multi-agent systems.

30

Subagent

A specialized agent called by the orchestrator to handle a specific subtask.

31

Multi-agent system

Multiple specialized agents collaborating on a task — each owns a narrow domain.

32

Handoff

Passing control and context from one agent to another mid-task.

33

Supervisor pattern

One agent manages and delegates to others — the most common enterprise multi-agent pattern.

34

Parallel execution

Multiple agents running simultaneously on different subtasks — cuts wall-clock time dramatically.

35

Agent loop

The continuous perceive → think → act → observe cycle that runs until the goal is reached.





How We Measure Agents

36

Trajectory

The full sequence of steps an agent took — essential for debugging and auditing.

37

Task completion rate

% of tasks fully completed without error — the primary production success metric.

38

Hallucination rate

How often the agent fabricates facts not grounded in tools or context.

39

Tool call accuracy

Correctness of tool selection and argument construction per agent step.

40

Latency

Time from user input to final agent response — p50/p95 matter most in production.

41

Token efficiency

Output quality per token used — directly maps to cost per agent run.

42

Human-in-the-loop

Checkpoint where a human reviews before the agent proceeds — critical for high-stakes tasks.





Keeping Agents Safe

43

Guardrails

Rules that prevent harmful, off-topic, or policy-violating outputs from the agent.

44

Sandboxing

Running agent code in an isolated environment — contains damage if something goes wrong.

45

Input validation

Checking user input before the agent processes it — blocks prompt injection and abuse.

46

Output filtering

Screening agent responses before delivery to catch harmful or sensitive content.

47

Prompt injection

Attack where malicious input in retrieved data hijacks agent behavior — top threat in 2026.

48

Rate limiting

Capping agent actions per minute to prevent runaway loops and cost explosions.

49

Kill switch

Hard stop that halts agent execution immediately — non-negotiable for production agents.





Tools Engineers Actually Use

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LangGraph

Graph-based agent orchestration — best for complex stateful workflows with branching logic.

51

CrewAI

Multi-agent framework with role-based collaboration — agents assigned as crew members.

52

AutoGen

Microsoft's conversational multi-agent system — great for code generation workflows.

53

Claude Code

Agentic coding assistant with file access, tool use, and long-horizon task completion.

54

LlamaIndex

Data framework for RAG and agentic workflows — strong on data connectors and retrieval.

55

Semantic Kernel

Microsoft SDK for LLM-powered pipelines — preferred in enterprise .NET environments.

56

OpenAI Assistants

Managed agent runtime with threads, memory, and tool use built in — fastest to prototype.





Shipping Agents That Work

57

Observability

Tracing every agent step end-to-end — you cannot debug what you cannot see.

58

Retry logic

Automatically retrying failed tool calls with backoff — prevents single-point failures.

59

Fallback

Default action when the agent fails or is uncertain — prevents blank/error responses.

60

Context window mgmt

Keeping prompt size within model limits across long multi-step agent runs.

61

Streaming

Returning agent output token-by-token as generated — makes agents feel responsive.

62

Cost tracking

Monitoring token spend per agent run — unchecked agents can burn through budgets fast.

63

Versioning

Tracking which agent config, prompt, and model produced which output — non-negotiable



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All 60 concepts. One playbook.

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shipping agents in 2026.

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